

SafetySense

George Teifel · Mary Teifel

UNM & CNM APP CONTEST 2026

WHO WE ARE

SafetySense calls for help when you can't.

George Teifel

University of New Mexico

Senior · Electrical Engineering

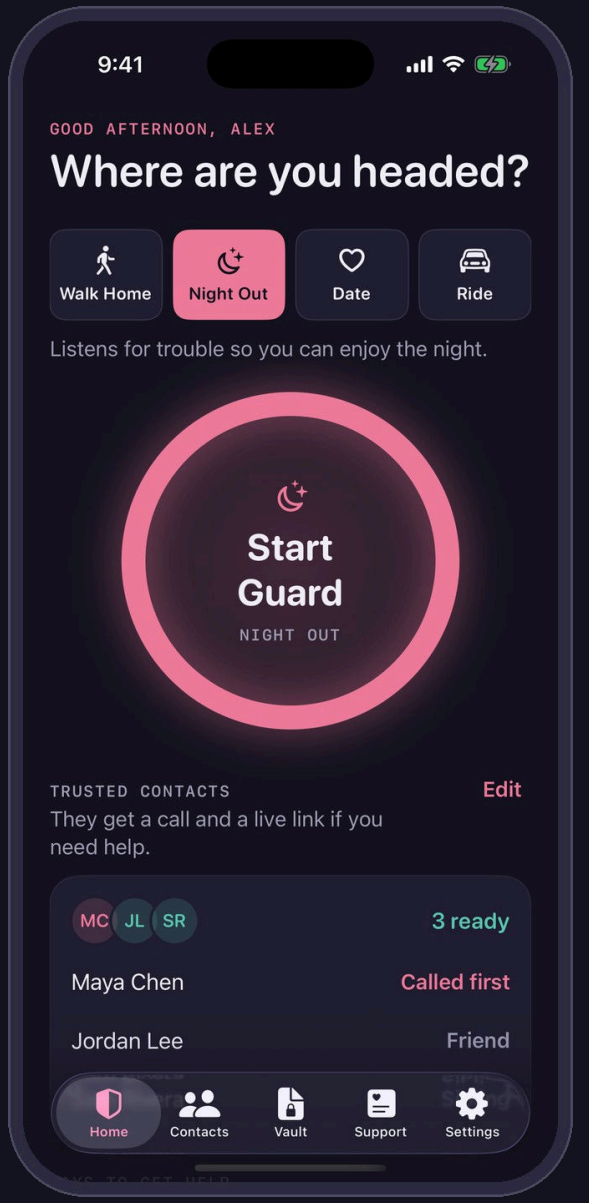
Built the iOS app and the backend

Mary Teifel

Central New Mexico Community College

Dual credit student · Mechanical Engineering

Research, design and the app's wording



THE PROBLEM

Most safety apps wait for you to press a button or set a timer

70%

of 298 women reported significant tonic immobility, an involuntary freeze, during sexual assault

Möller et al., Acta Obstet Gynecol Scand, 2017 (n = 298)

Freezing is involuntary

It is a normal response, not a choice.

Phones get taken, broken or switched off

A dead phone sends nothing.

Speaking isn't always possible

Even texting 911 takes a phone in your hand.

What if you can't press the button?

“Most safety apps”: based on the documentation of the apps we reviewed. We did not test them.

OUR ANSWER

You turn on a Guard. If you can't respond, it acts for you.

START IT

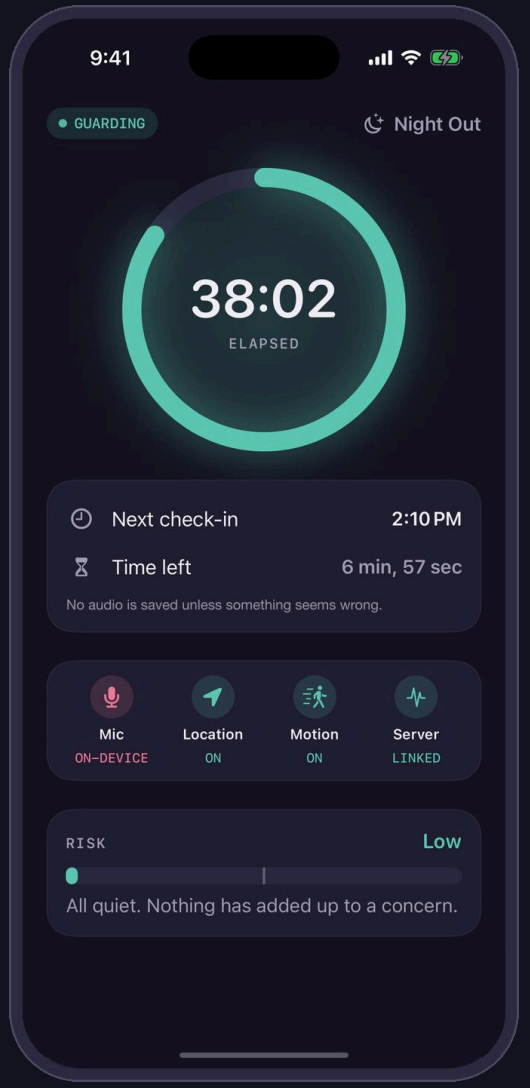
Before a walk home, a night out, a date or a ride, tap Start Guard.

Or skip straight to an alert with the Action Button or Siri.

WHAT IT WATCHES FOR

- **Screams**
- **Breaking glass**
- **Safe word**
- **Fall, then stillness**

It only listens while a Guard is on.



BUILT WITH

A native iPhone app · The listening happens on the phone
Real phone calls · Professional dispatch (test system today)

Watch three things.

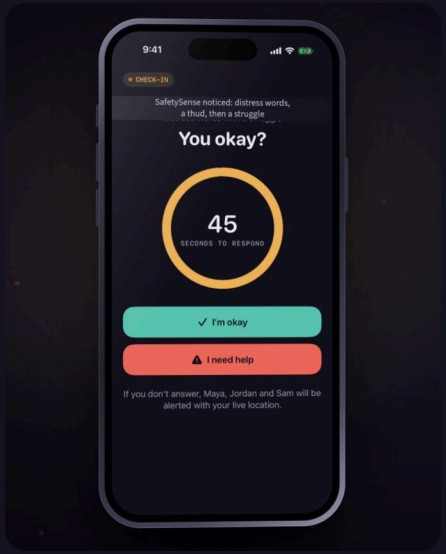
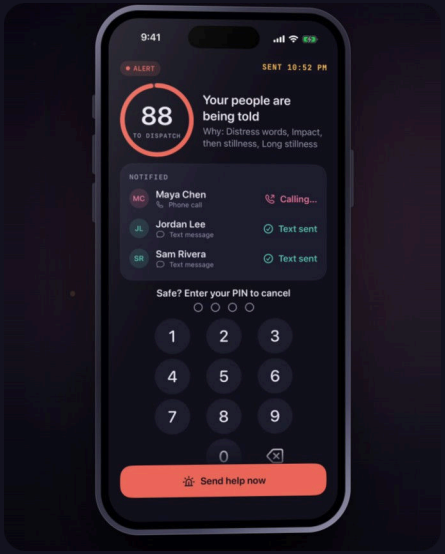
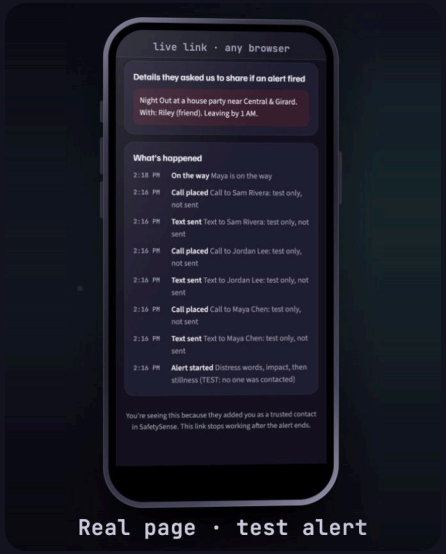
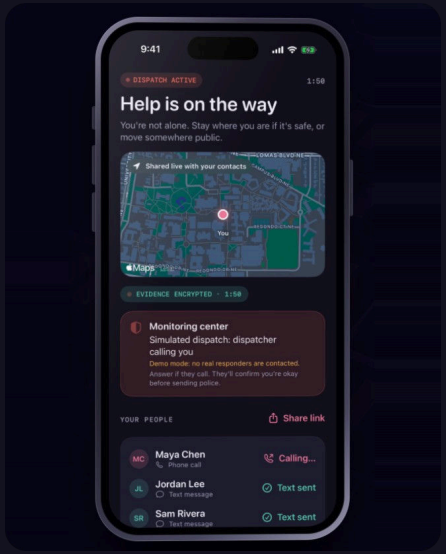
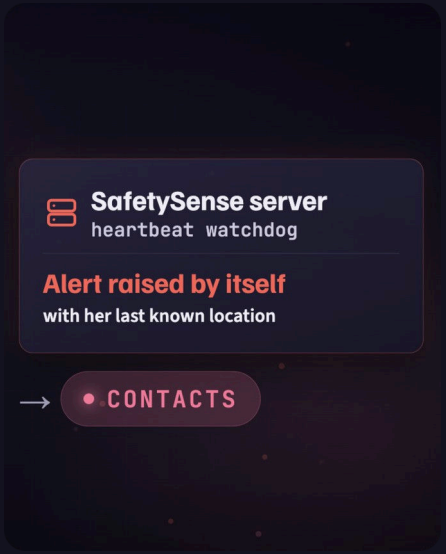
1. It asks before it alerts.
2. What her friend sees, and who's called if no one answers.
3. What happens if someone makes her cancel, or breaks her phone.





WHAT YOU JUST SAW

The app is real. Here's exactly what was simulated.

REAL APP	CALLS WORK	REAL PAGE	SIMULATED	ANIMATION
				
Recorded in the simulator, in demo mode	Real calls in our tests, not in this recording. Texts are sent but don't arrive yet.	Screenshot of a test alert	No police contacted. We use a test system	Of a check our server runs

Real page · test alert

Waits are sped up, and Ava and Maya are animated characters.
The duress PIN is built; what it reaches is the same simulated dispatch.

ORIGINALITY

Others have pieces. We connected the chain.

	UNM Safe	Noonlight	bSafe	Apple built-in	SafetySense
Notices distress on its own	–	Crash (paid)	Voice phrase	Crash; fall on Watch	✓ **Sound, motion, context**
Acts when the phone goes dark	–	Not documented	Not listed	Check In, timing varies	✓ **Our server notices**
Duress PIN (fake cancel)	–	–	Not listed	–	✓ **The two PINs you saw**
Professional dispatch	Call UNMPD	✓	–	911 call	✓ **Yes (test system)**
Live location to a friend	✓	✓	✓	✓ Check In	✓

From each product's documentation; not tested by us. These are four tools a student could already have.
Their columns are shipped products. Ours is a prototype; dispatch runs in a test system.

DESIGN AND EXPERIENCE

Designed for the worst night of someone's life

Calm words, never blame

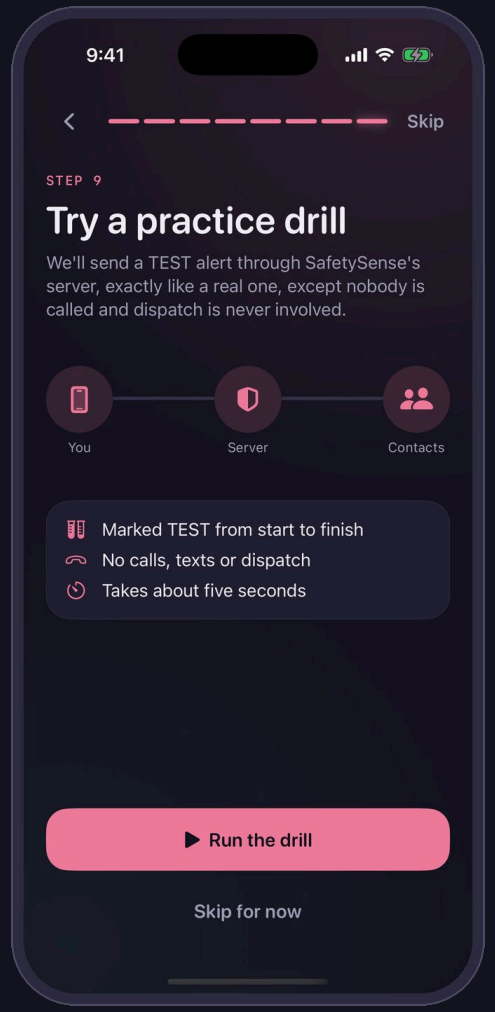
Written to be calm and never blaming.

Your first alert is a drill

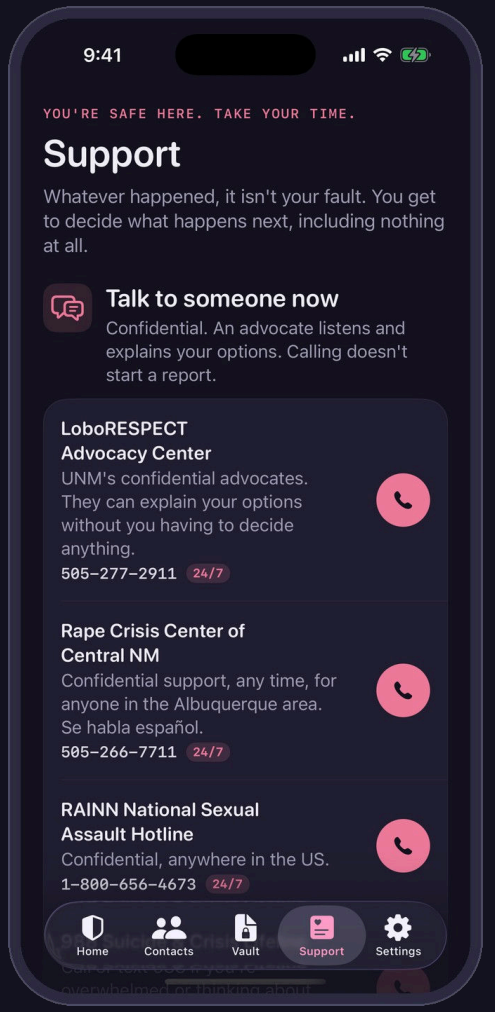
Onboarding ends with a practice run marked TEST.

Afterward, you choose

An advocate, medical care, reporting, or keeping it private.



The practice drill



Support, afterward

PRIVACY AND SECURITY

Private by default. Evidence only when it matters.

1

Guard on

Listening happens on the phone

2

Only the last 30 seconds

Kept in memory, then overwritten

3

Only if an alert fires

Audio is locked on the phone before upload

4

Vault

Opens with your Face ID. You decide who hears it

No audio is saved unless an alert fires

No ads, no trackers

Live links are built to expire after the alert

HOW IT'S BUILT

A real app, with a real server behind it

On the iPhone

- Listens for sounds and your safe word
- Feels for a fall, then stillness
- Clear rules decide when to act

heartbeat
+ location
no audio while guarding

Our server

- Runs the alert steps
- Notices if your phone goes silent

Reaching people

- Phone calls and texts to your contacts
- A web page for your friend
- Professional dispatch (test system today)

PATHWAY

How we'll know it works, and what we haven't proven yet

TARGETS, NOT YET MEASURED

< 1
false alert per 10 h

< 60 s
scream to ringing phone

< 8%
battery per hour

80%
finish onboarding

Not proven yet

- No field test in real, noisy places
- Real-device run on an iPhone is pending
- Live dispatch still needs our dispatch partner's sign-off
- A private test build, not public yet

NEXT

Student usability tests and advocate review → **Student pilot** → **Campus partnership**

SAFETYSENSE

When you can't call for help, SafetySense does.

SafetySense supplements 911. It never replaces it.

George Teifel UNM **Mary Teifel** CNM

Thank you. We'd love your questions.

